METHODS OF STUDYING THE CELL

Dr N. AILANE – First Year of Medicine - 2025/2026

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1 INTRODUCTION

Cellular study requires diverse methodologies for observation, characterization, and analysis. Microscopy, the foundational tool enabling cell discovery and the birth of cell biology, ranges from optical to electron microscopy. Combined with biochemical techniques like cell fractionation, these complementary methods reveal cellular morphology, ultrastructure, and molecular composition at different organizational levels.

2 MICROSCOPY

2.1 FUNDAMENTAL MICROSCOPE PARAMETERS

All microscopes are characterized by two essential parameters:

Magnification: How many times the final image has been enlarged.

Resolving Power: is the critical parameter conditioning image clarity. It's inversely proportional to the Resolution which is the minimum distance between two points that can be distinguished as separate entities. The human eye has a resolution of approximately 200 µm. Resolution defines how small two details can be and still be seen separately while Resolving Power determines how good is the system at distinguishing fine details.

2.2 LIGHT MICROSCOPY

2.2.1 STRUCTURAL COMPONENTS

Mechanical system: Facilitates manipulation and focusing().

Optical system: Generates the image (light source, condenser with diaphragm, objectives, eyepieces also called ocular lenses).

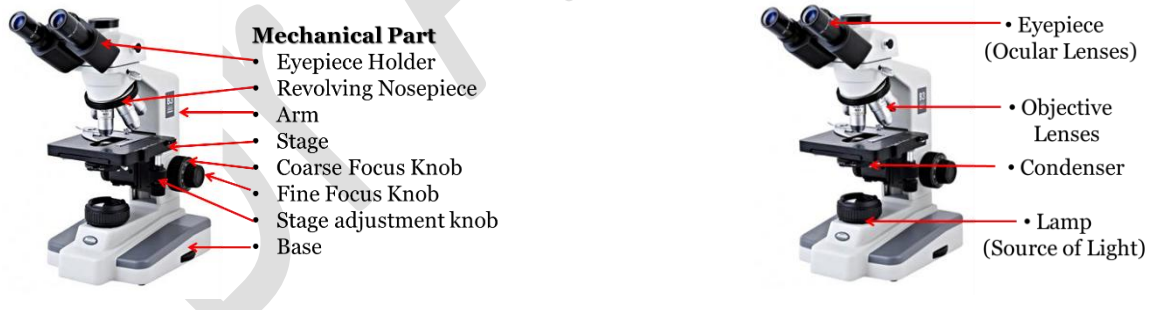


Figure 1 Structure of Light Microscope

2.2.2 LIGHT PATH AND IMAGE CONSTRUCTION

In a light microscope, the image is formed by the transmission of photons through a system of superimposed glass lenses. Light (photons) passes through the specimen and is refracted by a series of lenses (objectives and eyepieces), producing a magnified image on the retina or image capture system.

2.2.3 SPECIFICATIONS

- Total magnification equals the product of the objective and ocular magnifications, with a maximum useful value of approximately ×1000.

- Resolution: ~0.2 µm

2.2.4 SPECIMEN PREPARATION

Samples require proper preparation including fixation, sectioning, and staining to enhance contrast and preserve cellular architecture (detailed protocols covered in Embryology Lab Session).

2.2.5 APPLICATIONS

Histological tissue examination, cytological cell identification, microbiological organism observation, cellular structure analysis, and medical diagnosis.

2.2.6 VARIANTS

Sometimes, cell components are invisible under the light microscope, they are transparent, colorless yet ordinary light microscopy cannot distinguish organelles or proteins. **Fluorescence Microscopy** helps us reveal them through specific labeling using fluorescence.

2.2.6.1 PRINCIPLE OF FLUORESCENCE

Fluorescence is the property of certain substances (fluorochromes) to absorb light at a specific wavelength and re-emit it at a longer wavelength. Each fluorochrome possesses characteristic absorption and emission spectra.

2.2.6.2 INSTRUMENTATION

The fluorescence microscope uses a white light source for transmission observation plus a system that separates white light into its spectrum. It includes an excitation filter that selects the incident wavelength that excites the fluorochrome.

2.2.6.3 COMMON FLUOROCHROMES

- FITC (Fluorescein)/GFP (Green Fluorescent Protein): Blue excitation, green emission for protein labeling
- Rhodamine: Green excitation, red-orange emission for mitochondrial labeling
- DAPI: UV excitation, blue emission for DNA binding

2.2.6.4 SPECIMEN PREPARATION FOR FLUORESCENCE MICROSCOPY (IMMUNOFLUORESCENCE)

2.2.6.4.1 PRINCIPLE

Antibodies conjugated to fluorochromes enable specific protein detection and localization.

2.2.6.4.2 TECHNIQUES

Direct immunofluorescence: Fluorochrome directly attached to primary antibody

Indirect immunofluorescence: Fluorochrome attached to secondary antibody

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2.2.6.4.3 SAMPLE TYPES

- Suspension cells (analyzed by flow cytometry – Refer to Cell Culture Chapter)
- Adherent cells (analyzed by fluorescence microscopy)

2.2.6.4.4 APPLICATIONS

Protein localization, co-localization studies, expression quantification, clinical diagnosis.

2.2.6.5 ADVANCED DESIGN: CONFOCAL FLUORESCENCE MICROSCOPE

This is an improved version of the fluorescence microscope. The light source is a LASER that scans the specimen point by point. Through computer processing, acquisitions are superimposed and the image is reconstructed, producing a sharp image that can even be rendered in 3D.

2.3 ELECTRON MICROSCOPY

2.3.1 OPERATING PRINCIPLE

Electron microscopes replace photon beams with electron beams emitted under vacuum, accelerated by high voltage, and focused by electromagnetic lenses. Images form on fluorescent screens.

There are two types of Electron Microscopes:

Transmission Electron Microscope (TEM): Provides ultrastructure

Scanning Electron Microscope (SEM): Enables 3D morphological study and reveals surfaces

2.3.2 PERFORMANCE SPECIFICATIONS

- Magnification: ranging from x50,000 For TEM to Up to ×1,000,000 for SEM
- Resolution: 0.2 nm (2 Å)
- Revolutionary achievement: Revealed cellular ultrastructure and enabled organelle discovery
- Critical Limitation: Living specimens cannot be observed.

2.3.3 TRANSMISSION ELECTRON MICROSCOPE (TEM)

2.3.3.1 MECHANISM

Electron beam penetrates through ultra-thin specimens, providing ultrastructural information about cells and organelles.

2.3.3.2 APPLICATIONS

Ultrastructural analysis, organelle architecture, viral structure examination, membrane system investigation, subcellular component characterization.

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2.3.4 SCANNING ELECTRON MICROSCOPE (SEM)

2.3.4.1 MECHANISM

Electron beam scans specimen surfaces without penetration. Reflected electrons produce three-dimensional surface images. Samples are coated with conductive materials (gold or carbon) to optimize electron reflection.

2.3.4.2 APPLICATIONS

Three-dimensional morphology, surface topography analysis, external cellular surfaces, tissue architecture, materials science.

3 BIOCHEMICAL ANALYSIS METHODS: CELL FRACTIONATION

The objective of this technique is to isolate cellular structures for individual biochemical analysis through homogenization followed by fractionation using ultracentrifugation.

3.1 HOMOGENIZATION

This step allows the disruption of plasma membranes to release cellular contents, producing a homogenate containing all cellular components.

3.1.1 ESSENTIAL CONDITIONS

Homogenization must be performed under rigorous conditions to preserve cell components:

- Isotonic buffer solutions
- Low temperature (4°C)
- Neutral pH

3.1.2 HOMOGENIZATION METHODS

- Mechanical (grinding)
- Physical (pressure, sonication)
- Chemical (osmotic shock, detergents)
- Biochemical (enzymatic digestion)

NB: While most organelles remain intact, the endoplasmic reticulum (ER) and Golgi apparatus (GA) and cell membrane fragment and spontaneously reseal into small vesicles known as microsomes.

3.2 FRACTIONATION BY ULTRACENTRIFUGATION

Cell fractionation employs centrifugal force to separate cellular components. High-speed centrifuges (>15,000 rpm) are termed ultracentrifuges.

3.2.1 DIFFERENTIAL ULTRACENTRIFUGATION (DUC)

Sequential centrifugation at progressively increasing speeds separates components by size (mainly) and density.

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3.2.1.1 PROCEDURE

Each centrifugation step sediments specific organelles into pellets, while lighter components remain in the supernatant for subsequent centrifugation.

The organelles sediment in the following order: nuclei, mitochondria together with lysosomes and peroxisomes, microsomes than ribosomes.

3.2.1.2 ADVANTAGES VS DISADVANTAGE

Simple methodology with large sample volumes, however incomplete purification.

3.2.2 DENSITY GRADIENT ULTRACENTRIFUGATION (DGU)

Separates components by buoyant density using density gradients (typically sucrose gradients of varying concentrations).

3.2.2.1 PROCEDURE

Homogenate is layered onto a pre-formed density gradient and centrifuged. Components migrate to gradient levels matching their specific density.

Density-Based Separation (top to bottom):

- Low density: Lipid droplets, lipoproteins

- Intermediate density: Membrane fragments, microsomes, Golgi, lysosomes, peroxisomes, mitochondria

- High density: Nuclei, ribosomes

3.2.2.2 ADVANTAGES VS DISADVANTAGE

Provide high-purity fractions in single centrifugation with excellent separation but limited to small sample volumes

4 METABOLIC STUDIES: PULSE-CHASE TECHNIQUE

Radioactive Isotopes are used to track metabolic pathways.
An isotope is a chemical element that differs from the natural element by its nuclear mass. It can be unstable and undergo disintegrations releasing electrons or radiation: radioactive isotope such as ^{3}H (tritium), ^{14}C, ^{32}P, ^{35}S, ^{22}Na, ^{42}K, and ^{125}I.

4.1 EXPERIMENTATION PROTOCOL

4.1.1 PULSE PHASE

Living cells are exposed to a radioactive metabolic precursor (called "hot") for a short duration. This precursor enters the cell and is incorporated into certain metabolic reactions.

4.1.2 CHASE PHASE

Cells are maintained in a medium containing a large excess of non-radioactive precursor (called "cold") which drive the cell to incorporate the cold precursor. Therefore, the use of hot precursor is accelerated in various metabolic pathways.

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4.1.3 ANALYSIS

Regular sampling at spaced intervals allows tracking transformations of the hot precursor into various molecules carrying radioactive labeling. The compartments containing radioactive molecules can de detected with the help of autoradiography using optical or electron microscopes.

4.1.4 APPLICATIONS

Protein synthesis and trafficking, DNA replication kinetics, secretory pathway analysis, metabolic transformation studies.

5 CONCLUSION

Modern cell biology employs integrated methodological approaches combining microscopic visualization with biochemical and metabolic analyses. Mastery of these fundamental techniques—from basic light microscopy to sophisticated fractionation and radiotracer methods—provides essential skills for understanding cellular structure, function, and dynamics in biomedical research and clinical practice.

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